

Underwater Object Detection Using Improved YOLOv8 Model Based on Efficient Multi-Scale Adaptive Fusion

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Abstract—This paper presents an enhanced YOLOv8 model tailored for underwater object detection, specifically addressing the challenge of poor detection performance in blurry underwater image scenes. Our approach introduces three key innovations: (1) a multi-scale adaptive fusion module based on cross-spatial learning, designed to refine and enhance feature representations by effectively filtering out less useful features; (2) a gated low-level detection module that improves feature extraction for small objects while controlling computational efficiency; and (3) a novel dynamic non-monotonic loss function, termed Wise-IoU, which leverages a focusing mechanism to boost the model's ability to handle complex underwater scenarios. Extensive experiments demonstrate that our improved model achieves superior performance on underwater object detection datasets. Additionally, we introduce a specialized dataset focused on underwater garbage object detection, aimed at advancing research into the persistent and non-decomposable challenges posed by industrial and household waste in aquatic environments.

Keywords—Underwater Object Detection, Multi-Scale Adaptive Fusion, Gated Low-Level Detection, Wise-IoU Loss, YOLOv8.

I. INTRODUCTION

The effective development and utilization of marine resources are crucial for improving people's livelihoods and promoting socio-economic development [1]. Traditionally, the collection of underwater resources relied heavily on manual operations, which pose significant risks to human safety due to the complex, dynamic nature of underwater environments characterized by increasing pressure, limited light, and low temperatures. These challenges underscore the need for alternative methods to explore and harness marine resources safely.

In recent years, underwater robots have emerged as a promising solution, offering new avenues for marine exploration. With the growing demand for seafood such as sea urchins, sea cucumbers, starfish, and scallops, there is an urgent need to advance underwater biological fishing technology. By deploying underwater robots instead of manual operations, we can enhance safety, improve

efficiency, and increase economic benefits in fishing activities [2].

However, underwater environments present unique challenges for object detection. Due to the presence of suspended particles in water and the susceptibility of light to absorption and scattering, underwater images often suffer from degraded quality. Developing an advanced object detection system optimized for underwater environments is essential for accurate identification of marine life, ensuring the efficient operation of underwater robots and enhancing safety in fishing operations.

Underwater object detection faces unique challenges compared to traditional tasks, including variable lighting conditions, turbidity, and complex backgrounds [3]. Addressing these challenges requires the development of robust underwater detection methods capable of overcoming such adversities.

This paper addresses the challenges of underwater object detection by proposing an enhanced YOLOv8 model designed to operate effectively in marine environments. Our enhanced model incorporates a multi-scale adaptive fusion module leveraging cross-space learning and weighted averaging techniques to effectively capture and utilize discriminative features. Additionally, a gated mechanism is employed in the low-level detection module to enhance feature extraction capabilities for small objects without compromising computational efficiency. Furthermore, we introduce the Wise-IoU loss function, a dynamic non-monotonic focusing mechanism that enhances the model's ability to handle complex underwater scenarios.

By integrating these advanced modules, the proposed YOLOv8-based framework demonstrates significantly improved performance in underwater object detection, thereby facilitating safer and more efficient marine resource exploration. The main contributions of this paper are summarized as follows:

(1) We propose a multi-scale adaptive fusion module based on cross-spatial learning and weighted averaging, which effectively enhances feature representation by suppressing background interference and preserving salient object features in underwater environments.

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(2) We design a gated low-level detection module to improve the detection of small underwater objects while maintaining a balance between detection accuracy and computational efficiency.

(3) We introduce a novel loss function, Wise-IoU, which employs a dynamic non-monotonic focusing mechanism to optimize bounding box regression, thereby improving model robustness and detection accuracy in complex underwater scenarios.

II. RELATED WORKS

A. Underwater Object Detection

Object detection in underwater environments presents unique challenges compared to terrestrial settings, including environmental complexity, limited data availability, and diverse target characteristics. These factors render underwater object detection a particularly demanding research endeavor. With advancements in ocean exploration technologies, the study of underwater target detection has garnered increasing attention from researchers.

Recent years have witnessed significant progress in object detection based on underwater imagery. Jalal *et al.* demonstrated the effectiveness of the YOLO model for detecting and classifying fish species by leveraging shape and texture features, complemented by Gaussian mixture models and optical flow algorithms to incorporate temporal information for identifying free-swimming fish in low-visibility conditions [4]. Addressing the challenges of complex environmental variations and limited training data, Zeng *et al.* proposed integrating adversarial occlusion networks into the Faster R-CNN framework [5], thereby enhancing the detection network's robustness by mitigating overfitting to fixed features [6].

Cai *et al.* advanced underwater object detection by employing an SSD framework that combines multi-scale features and incorporates attention mechanisms, significantly improving the detectability of small objects within the URPC dataset [7]. Qiang *et al.* introduced deep separation deformable convolution to address the dual challenges of low accuracy and poor real-time performance in complex underwater environments [8]. Li *et al.* developed a real-time detection algorithm for underwater fish targets in uncontrolled environments, building upon an improved YOLO architecture and transfer learning techniques. Their

approach employs adaptive histogram equalization for data preprocessing and demonstrates superior performance for small and overlapping objects in aquatic settings [9].

Xin *et al.* proposed a lightweight underwater object detection method, YOLOv7-SDBB, designed to tackle the prevalent issues of complex backgrounds and target overlap in underwater environments [10]. These advancements collectively underscore the growing sophistication and diversity of approaches in addressing the unique challenges of underwater object detection.

III. PROPOSED METHODS

A. Baseline Model

The YOLOv8 model is selected as our baseline framework due to its enhanced efficiency and speed compared to its predecessor, the YOLOv5 model. Building upon YOLOv5, YOLOv8 introduces several architectural improvements that make it particularly suitable for our application. In the Backbone section, YOLOv8 integrates the CBS module, C2f module, and SPPF module, drawing inspiration from the ELAN design philosophy of YOLOv7. Notably, the C3 module from YOLOv5 has been replaced with a more lightweight and gradient-rich C2f module. This modification not only enhances computational efficiency but also allows for dynamic channel adjustment across different model scales, enabling better adaptability to object detection tasks of varying complexity. While both YOLOv8 and YOLOv5 utilize the PAN-FPN structure, YOLOv8 further refines this by removing the convolutional layers in the upsampling section and incorporating the C2f module. In the Head section, YOLOv8 adopts a decoupled head structure, separating the classification and detection heads to improve model flexibility and scalability. Additionally, YOLOv8 transitions from an Anchor-Based approach to an Anchor-Free method, eliminating the dependency on anchor boxes for improved simplicity and performance.

To further enhance detection accuracy and model stability, YOLOv8 introduces two key innovations: the TaskAligned Assistant positive sample allocation strategy and Distribution Focal Loss as objective functions. These additions enable dynamic adjustment of positive and negative sample weights during training, addressing class imbalance issues and improving overall detection performance.

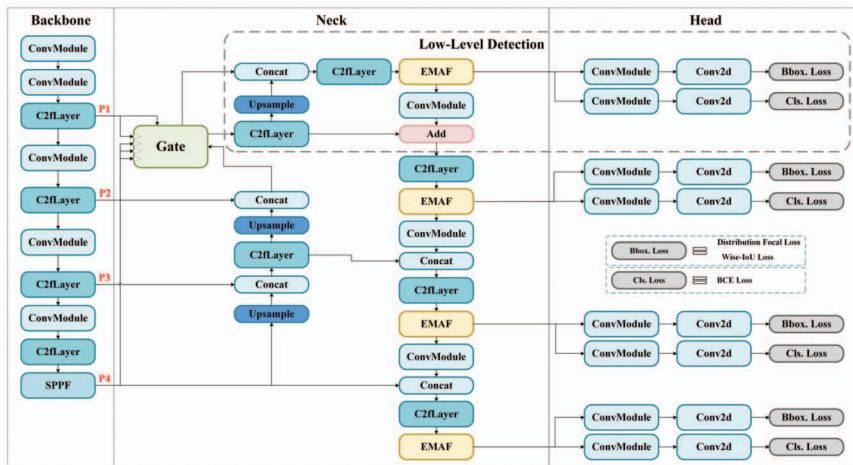


Fig. 1. The structure of improved YOLOv8 model.

Considering these advancements, this work develops an underwater object detection model based on the YOLOv8 framework, leveraging its improved efficiency, scalability, and robustness for enhanced performance in complex aquatic environments.

B. Improved YOLOv8 Model

We present an improved YOLOv8 model for underwater object detection, which introduces efficient multi-scale adaptive fusion (EMAF) module, gated low-level detection (GLD) module and Wise-IoU loss. The structure of our model is shown in Fig. 1, and the detailed improvement is shown in the following content.

C. Efficient Multi-Scale Adaptive Fusion Module

We propose an Efficient Multi-scale Adaptive Fusion (EMAF) module, whose architecture is detailed in Fig. 2. This module processes feature maps by partitioning them into multiple sub-feature groups and effectively capturing spatial information across different scales through a parallel sub-network structure. Within each sub-feature group, the EMAF module dynamically recalibrates channel weights and establishes pixel-level pairwise relationships via global information encoding and cross-dimensional interactions. This design significantly enhances the model's feature representation capabilities while maintaining computational efficiency—introducing only a minimal increase in parameters and computational overhead.

To further optimize computational efficiency, the EMAF module employs a novel strategy: remapping a subset of channels to the batch dimension while dividing the channel dimension into multiple sub-feature groups. This approach ensures preservation of channel-specific information while simultaneously reducing computational complexity. Consequently, our model achieves superior performance without incurring excessive computational costs or memory burden.

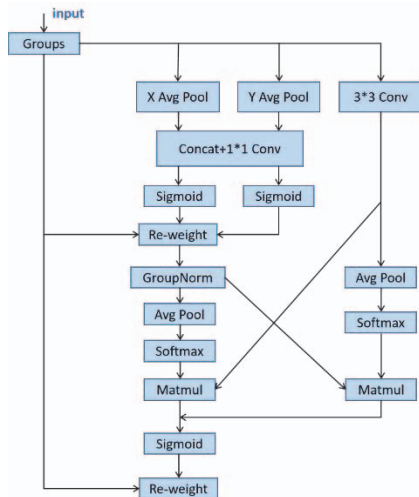


Fig. 2. The structure of efficient multi-scale adaptive fusion module.

Moreover, within each parallel branch, the EMAF module dynamically adjusts channel weights through global information encoding and leverages cross-dimensional interactions to capture pixel-level correlations. These advancements enable the EMAF module to significantly

enhance object detection accuracy on challenging underwater datasets. Our experimental results demonstrate that the proposed method achieves state-of-the-art performance in submerged environments while maintaining efficient computational requirements.

D. Gated Low-Level Detection Module

As shown in Fig. 3, approximately 10% of the objects in the URPC dataset are small targets. To address this challenge, we propose a gated low-level detection module that effectively balances accuracy and efficiency.

The proposed module introduces additional foreground (object) and background information to improve detection performance for small objects. Specifically, we incorporate a low-level feature map to complement the higher-level feature maps through fusion. This allows the model to better capture fine details critical for detecting small targets.

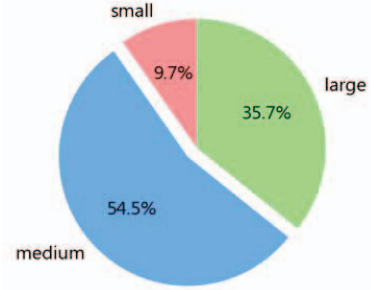


Fig. 3. distribution of object size in URPC dataset.

To mitigate the computational overhead introduced by incorporating extra feature maps, we draw inspiration from the Mixture of Experts (MoE) method and design a gated mechanism. The module takes P1, P2, P3, and P4 feature maps as inputs and generates control signals through max pooling and fully connected layers. These signals enable the model to dynamically decide whether to activate the small object detection pathway, thereby optimizing average inference speed while maintaining detection accuracy.

E. Wise-IoU Loss

Intersection over Union (IoU) is a critical metric for evaluating the similarity between predicted and ground-truth values. It represents the ratio of the overlapping area between the predicted bounding box and the ground-truth bounding box to their total area. A higher IoU value indicates that the predicted structure is closer to the ground truth, reflecting better model performance.

The loss function measures the difference between predicted and ground-truth values. In the YOLOv8 model, VFL (Varifocal Loss) is used as the classification loss, while DFL (Distribution Focal Loss) combined with IoU loss is employed for regression tasks [11]. The definition of CIoU Loss is provided in Eq. (1):

$$L_{CIoU} = 1 - IoU + \frac{d^2}{c^2} + \alpha v \quad (1)$$

In the formula, d represents the distance between the center points of the predicted box and the ground-truth box, c is the diagonal distance of the smallest bounding rectangle. α is the smoothing coefficient, and v represents the relative value of the aspect ratio of the box.

CIoU Loss incorporates shape information and diagonal distance into its design and introduces a correction factor to enhance model detection accuracy and robustness, particularly in complex scenes. However, CIoU Loss has some limitations: it requires additional information, is sensitive to parameter settings, and demands careful tuning of hyperparameters. Furthermore, parameter uncertainty can interfere with prediction results. In underwater image datasets, target features are often less distinct, with high background similarity and dense object stacking, leading to lower annotation quality and degraded model detection performance.

To address these issues, Wise-IoU Loss was proposed [12]. It introduces a dynamic non-monotonic focusing mechanism that allocates gradient gains based on the "outlier degree" of anchor boxes. This mechanism reduces the model's reliance on high-quality anchor boxes while minimizing harmful gradients from low-quality examples, effectively improving classification accuracy. Wise-IoU Loss is divided into three versions: WIoUv1, WIoUv2, and WIoUv3, with their respective calculation methods shown in Eqs. (2), (3), and (4).

WIoUv1 calculates the scaling weight based on distance measurement to control IoU loss. WIoUv2 introduces the thought of focal loss [13] to make model more focus on difficult instances. WIoUv3 constructs the gradient gain controlling method through non-monotonic focusing coefficient β to control the gradient gain from box regression.

$$L_{WIoUv1} = \exp\left(\frac{(x-\tilde{x})^2 + (y-\tilde{y})^2}{W_g^2 + H_g^2}\right)(1 - IoU) \quad (2)$$

$$L_{WIoUv2} = (1 - IoU)^\gamma L_{WIoUv1} \quad (3)$$

$$L_{WIoUv3} = \left(\frac{\beta}{\delta\alpha\beta - \delta}\right) L_{WIoUv1}, \beta = \frac{(1 - IoU)^\gamma}{1 - IoU} \quad (4)$$

In the Exp. (2), $\tilde{x}$ and $\tilde{y}$ represent the coordinates of the ground-truth object. x and y represent the coordinates of the predicted objects. In the Exp. (3), $\gamma \in (0, +\infty)$ is a super parameter to control curve curvature. In the Exp. (4), α and β are super parameters to control the gradient gain.

Through systematic experiments, WIoUv3 demonstrated superior performance in underwater object detection tasks, achieving higher precision and recall compared to its predecessors. The dynamic non-monotonic focusing mechanism effectively addresses the challenges of ambiguous annotations and complex scenes in underwater environments.

IV. EXPERIMENTS

A. Experimental Configuration & Evaluation Metrics

The experiments were conducted using an NVIDIA RTX 3090 with PyTorch 1.10 and CUDA 11.3 as the experimental environment. During the training process, the number of workers was set to 32, with a batch size of 32. The AdamW optimizer was employed, with an initial learning rate of 0.00125, a momentum factor of 0.91, and 120 epochs. All other parameters were set to their default values.

To evaluate our model's performance, we utilized several metrics: precision (P), recall (R), average precision (AP), and mean average precision (mAP). These metrics provide a

comprehensive assessment of the model's accuracy and effectiveness in object detection tasks.

B. Datasets

URPC2020 Dataset: This open-source dataset was used in the Underwater Object Detection Competition - Optical Competition, comprising four main categories: sea cucumber, sea urchin, starfish, and scallop. It contains a total of 5,543 images. During our experiments, we randomly divided the dataset into training, validation, and test sets in an 8:1:1 ratio. Additionally, a small number of seaweed labels were removed to focus solely on the primary categories of interest.

Our Dataset: To further validate the model's generalization capability and adapt it to real-world scenarios involving underwater robots, we captured images using an RGB camera mounted on an underwater robot. This dataset includes 813 images across eight categories: plastic bottles, plastic bags, cardboard boxes, paper cups, cans, wood, glass bottles, and Styrofoam pieces. These categories represent the most common types of underwater garbage. During our experiments, the training and validation sets were randomly divided in an 8:2 ratio to ensure a robust evaluation.

C. Ablation Experiment

To demonstrate the effectiveness of each module, the ablation experiment results are shown in Tab. I. The results showed that the addition of each module improved the performance of the model. Among them, the addition of WIoU alone was more effective in improving mAP, increasing it by 0.005. On this basis, the addition of detect improved the accuracy by 0.24, showing a significant improvement. After adding EMA module, although the accuracy decreased slightly, it still improved by 0.11 compared to the original model, while other indicators improved, indicating the effectiveness of the experiment.

TABLE I. THE RESULTS OF ABLATION EXPERIMENT

EMAF	GLD	WIoU	P (%)	R (%)	AP ₅₀ (%)	mAP (%)
-	-	-	82.0	75.4	83.6	49.9
$\mathcal{R}$	-	-	83.8	75.1	83.6	48.7
-	$\mathcal{R}$	-	82.7	75.2	83.4	49.1
-	-	$\mathcal{R}$	83.1	76.2	83.8	49.8
$\mathcal{R}$	$\mathcal{R}$	-	83.3	76.1	83.6	49.4
$\mathcal{R}$	-	$\mathcal{R}$	83.5	75.8	83.6	49.1
-	$\mathcal{R}$	$\mathcal{R}$	84.4	75.5	83.8	49.7
$\mathcal{R}$	$\mathcal{R}$	$\mathcal{R}$	84.1	76.3	84.1	50.3

EMAF Module: We explore the impact of adding EMAF module at different positions as shown in Fig. 4, and Tab. II investigates the results at several positions. The structure indicates that adding EMAF module at position ③ has the best performance. The analysis is that the locations ① and ② are still in base feature extraction process, and the feature is unstable. Adding EMAF module would disturb this process.

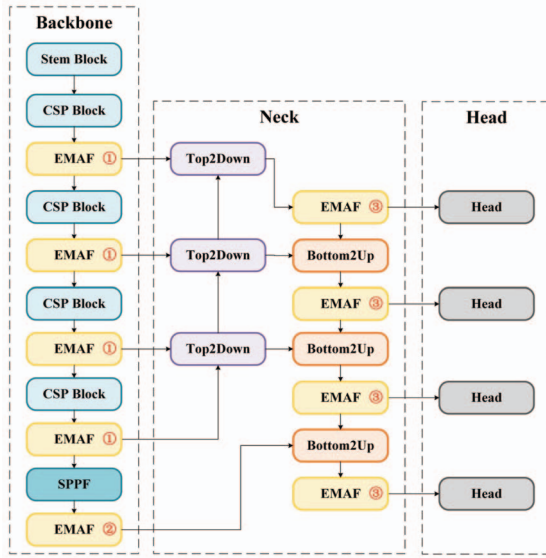


Fig. 4. The location of the EMAF module.

TABLE II. THE RESULTS OF EMAF MODULE

Location	P (%)	R (%)	AP ₅₀ (%)	mAP (%)
①	80.9	76.3	82.6	48.8
②	80.4	76.8	83.1	49.5
③	83.8	76.3	83.6	49.9
①+②	80.4	74.8	83.3	49.2
①+②+③	83.2	77.0	83.0	48.6

Wise-IoU Loss: Wise IoU loss is greatly affected by hyperparameters, and the parameter γ can effectively reduce the loss value of negative samples (simple samples). The higher the probability of simple samples, the stronger the effect. Choosing the appropriate gamma has a significant impact on improving model performance. To reflect the influence of different gamma values, a comparative experiment was conducted below, and the experimental results are shown in Tab. III. The results indicate that when γ is set as 1.7, Wise IoUv3 has the best effect, so it is used as an improvement method.

TABLE III. THE RESULTS OF WISE-IOU LOSS

Methods	P (%)	R (%)	AP ₅₀ (%)	mAP (%)
Wise-IoUv1	81.0	77.3	83.0	49.6
Wise-IoUv2($\gamma=0.5$)	83.1	74.4	83.6	49.4
Wise-IoUv2($\gamma=1.5$)	81.6	77.6	83.8	49.8
Wise-IoUv3($\gamma=0.5$)	81.5	75.9	83.4	49.8
Wise-IoUv3($\gamma=1.5$)	82.1	77.1	84.0	49.8
Wise-IoUv3($\gamma=1.7$)	84.1	77.6	84.1	49.8
Wise-IoUv3($\gamma=1.9$)	81.7	76.8	83.8	49.8

D. Comparison Experiments

URPC2020 Dataset: In order to further validate the performance of the improved model, a comparative experiment was conducted on the classic models in the YOLO series. The average accuracy value AP of four types in the URPC2020 dataset was selected as the comparison

index, and the results are shown in Tab. IV. Overall, all indicators of our model have increased, with sea urchins and starfish showing a more significant improvement in AP values, increasing by 0.09 and 0.1 respectively compared to YOLOv8, mAP50 increasing by 0.04, and YOLOv5 increasing by 0.45, while the parameter count only increased by 0.71M. It can be seen that the method used in this article is still quite effective.

TABLE IV. THE RESULTS OF URPC2020 DATASET

Models	AP(%)				mAP (%)
	Holothurian	Echinus	Scallop	starfish	
YOLOv5	63.6	90.7	78.3	85.6	79.5
YOLOv6	68.4	91.0	80.4	86.1	81.7
YOLOv7	73.3	92.2	82.1	87.0	83.6
YOLOv9	73.2	91.4	82.3	87.4	83.6
YOLOv10	73.0	91.1	82.2	87.2	83.3
YOLOv11	73.5	91.5	82.1	87.4	83.4
Our	74.6	92.3	82.3	88.4	84.0

Our Dataset: To verify the generalization ability of the model, different models were compared on the underwater garbage dataset, as shown in Tab. V. The results indicate that the improved model still has certain advantages compared to the classical model on different datasets, with an average accuracy of 0.893, ranking second among the compared models and an improvement of 0.15 compared to the original model. Although it is lower than the highest YOLOv7, it still has an advantage in parameter quantity, indicating that the improved model has a certain degree of generalization ability.

TABLE V. THE RESULTS OF OUR DATASET

Models	P(%)	R(%)	AP ₅₀ (%)	mAP(%)
YOLOv5	90.9	76.0	83.8	78.5
YOLOv6	89.9	84.4	87.3	82.1
YOLOv7	83.7	84.8	89.3	84.2
YOLOv9	85.4	80.2	87.8	82.6
YOLOv10	87.2	78.9	85.6	80.1
YOLOv11	91.5	83.7	88.4	83.9
Our	93.4	85.6	90.6	85.5

E. Visualization

The following results compare the detection performance of our model and YOLOv8 on the URPC2020 dataset, and compare them with real values. The first column shows the original annotated images, the second column shows the detection results of YOLOv8, and the third column shows the detection results of our model. The results are shown in Fig. 5. The results showed that our model can detect targets well in different underwater environments. At the same time, it can be found that due to the high similarity between underwater organisms such as sea cucumbers and scallops and the background environment, it is easy to cause false detection problems. Moreover, the model before improvement had the problem of detecting the same target multiple times. After the model improvement, the phenomenon of the above problems has been reduced to a certain extent, and the confidence has been significantly

improved. This further proves that the improved model has better detection performance.

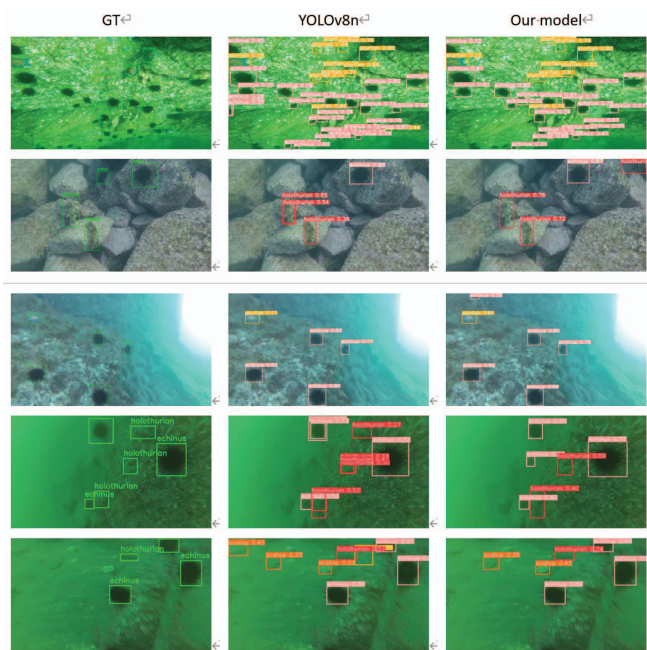


Fig. 5. The visualization comparison.

V. CONCLUSIONS AND FUTURE WORK

This paper presents an enhanced underwater object detection model based on YOLOv8, incorporating the efficient multi-scale adaptive fusion module, the gated low-level detection module, and the Wise-IoU loss function. Experimental results demonstrate that our approach achieves notable improvements over the baseline in detecting underwater targets. Despite these gains, several challenges remain. Future work will focus on further model lightweighting to support deployment on resource-constrained devices, enhancing detection performance for visually ambiguous categories such as sea cucumbers through targeted data augmentation, and optimizing the detection head to address issues arising from densely stacked underwater objects.

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